

Agricultural Science Waec past Questions Candidate/Private

Question 1

- (a) (i) Explain the term agriculture. [2 marks]
- (ii) List three industries that utilize cereals as their raw materials. [3 marks]
- (b) Outline six problems that hinder the attainment of self-sufficiency in food production in West Africa. [6 marks]
- (c) Enumerate five benefits of commercial farming. [5 marks]

Question 2

- (a) Explain the term farm mechanization. [2 marks]
- (b) List six limitations of farm mechanization in West Africa. [6 marks]
- (c) Suggest eight possible ways of encouraging the use of farm mechanization to improve agricultural production in West Africa. [8 marks]

Question 3

- (a) List two sources of carbon dioxide in the atmosphere. [2 marks]
- (b) (i) State four causes of soil acidity. [4 marks]
- (ii) State two ways by which soil pH influences crop growth. [2 marks]
- (c) List three ways by which each of the following biotic factors affects agricultural production:
 - (i) Parasites;
 - (ii) Soil organisms. [6 marks]
- (d) State two advantages of zero tillage. [2 marks]

Question 4

- (a) (i) Explain the term drainage. [2 marks]
- (ii) State four ways by which drainage is important in agriculture. [4 marks]
- (b) (i) List two methods of drainage. [2 marks]
- (ii) State two advantages and two disadvantages of each method you listed in 4(b)(i). [8 marks]

Question 5

- (a) What are ornamental plants? [2 marks]
- (b) List five factors that influence the choice of a crop nursery site. [5 marks]
- (c) Give two examples of each of the following methods of weed control:
 - (i) Physical method; [2 marks]
 - (ii) Biological method; [2 marks]
 - (iii) Cultural method. [2 marks]
- (d) Give three functions of potassium in crops. [3 marks]

Question 6

- (a) State four advantages of grafting. [4 marks]
- (b) Mention four ways by which forests enhance soil conservation. [4 marks]
- (c) State two forestry regulations in your country. [2 marks]
- (d) In an afforestation programme, teak seedlings were planted at 4m x 4m. After transplanting, 10.5% of the seedlings were lost to stress. Before harvesting, additional 12.8% were lost to wind damage. Calculate:
 - (i) The number of teak seedlings required to plant 2.3 hectares of farmland. [2 marks]
 - (ii) The number of teak seedlings lost to stress. [2 marks]
 - (iii) The number of teak stands harvested. [2 marks]

Question 7

- (a) What is fish farming? [2 marks]
- (b) Define the following terms as used in fish farming:
 - (i) Stocking rate; [1 mark]
 - (ii) Fingerlings; [1 mark]
 - (iii) Polyculture; [1 mark]
 - (iv) Fishing gears. [1 mark]
- (c) List four factors that influence the water requirements of farm animals. [4 marks]
- (d) Discuss foot-and-mouth disease under the following headings:
 - (i) Causal agent;
 - (ii) Symptoms;
 - (iii) Control. [6 marks]

Question 8

- (a) Draw and label the digestive system of a ruminant animal. [6 marks]
- (b) Describe the process of digestion in the ruminant animal. [10 marks]

Question 9

- (a) (i) List the four factors of agricultural production. [2 marks]
- (ii) State the reward for each factor you have listed in 9(a)(i). [2 marks]
- (b) Outline three functions of the wholesaler in agricultural marketing. [3 marks]
- (c) State two objectives of agricultural extension. [4 marks]
- (d) Mention five examples of group extension teaching methods. [5 marks]

Question 10

- (a) List four factors that affect the efficiency of agricultural labour. [4 marks]
- (b) Outline four ways in which capital is important in agricultural production. [4 marks]
- (c) List five agencies that are involved in agricultural extension programmes in West Africa. [5 marks]

(d) Give three reasons why governments in West Africa should encourage agricultural extension services. [3 marks]

Question 11

(a) List four qualities of a good extension agent. [4 marks]

(b) State two advantages and two disadvantages of each of the following agricultural extension teaching methods:

- Individual method
- Group method [8 marks]

(c) What are forest regulations? [2 marks]

(d) State four precautionary measures that should be taken in beekeeping. [4 marks]

Question 12

(a) List five non-governmental agricultural organizations in West Africa. [5 marks]

(b) State four ways in which agriculture and industry are interrelated. [4 marks]

(c) Give one function of each of the following parts of a disc plough:

- Disc
- Beam
- Furrow wheel
- Disc scraper
- Hitch [5 marks]

(d) State four disadvantages of animal power. [4 marks]

Question 13

(a) State five factors that should be considered in choosing an irrigation system in West Africa. [5 marks]

(b) Mention four problems associated with irrigation in West Africa. [4 marks]

(c) A rectangle-shaped farmland has a length of 0.4km and a breadth of 2,400m, calculate the area of the farmland in hectares. [5 marks]

(d) List one primary tillage and three secondary tillage implements. [4 marks]

Question 14

(a) Draw a well-labeled diagram of the water cycle. [8 marks]

(b) Mention two deficiency symptoms of phosphorus in crops. [2 marks]

(c) Using the table below, classify the following insect pests according to their modes of feeding:

- Weevil, Locust, Scale insect, Armyworm, Earworm, Cotton stainer, Green spidermites, Thrips, Capsids, Yam beetle. [10 marks]

Question 15

(a) Discuss maize smut disease under the following headings:

- Causal organism
- Mode of transmission
- Two symptoms
- Two cultural control measures [6 marks]

(b) Enumerate four forest management practices in West Africa. [4 marks]

(c) (i) State four classes of farm animals based on their uses. (ii) Give one example of each class stated in 4(c)(i). [8 marks]

Question 16

(a) Discuss anthrax disease under the following headings:

- Causal organism
- Mode of transmission
- Two symptoms
- Two prevention and control measures [6 marks]

(b) (i) List four sources of plant protein in livestock feeds. (ii) Give three examples of forage legumes that could be found in rangelands in West Africa. [7 marks]

(c) Mention five characteristics of labor as a factor of agricultural production. [5 marks]

Question 17

(a) Explain briefly each of the following basic economic concepts:

- Scarcity
- Scale of preference [4 marks]

(b) Give four reasons why agricultural insurance is important. [4 marks]

(c) State five fishery regulations in West Africa. [5 marks]

(d) Give five reasons why beekeeping is important. [5 marks]

Question 18

(a) Explain three effects of each of the following problems on agricultural development in West Africa:

- Unpredictable weather
- Inadequate rural infrastructure
- Disease [9 marks]

(b) Mention four examples of surface irrigation methods. [4 marks]

(c) State five problems associated with irrigation in West Africa. [5 marks]

Question 19

(a) State three ways in which each of the following factors limit agricultural mechanization in West Africa:

- Economic factor
- Technical know-how
- Small farm holdings [9 marks]

(b) Explain three classes of insect pests based on their feeding habits. [9 marks]

Question 20

(a) State four qualities of good draught animals. [4 marks]

(b) Name two examples of draught animals. [2 marks]

(c) State four advantages of animal power. [4 marks]

Question 21

(a) List six abiotic components of a farm ecosystem. [6 marks]

(b) State four ways in which the carbon cycle is important in agriculture. [4 marks]

(c) A vegetable crop is planted at a spacing of 40cm x 40cm on an 80m x 40m farmland

Question 21

(c) A vegetable crop is planted at a spacing of 40cm x 40cm on an 80m x 40m farmland. Calculate the expected population of the crops on the farmland. [3 marks]

(d) Name four examples each of crops that belong to the following crop groupings:

- Fibres
- Beverages [4 marks]

Question 22

(a) Name five parts of the digestive system of a chicken. [5 marks]

(b) Give one function of each of the parts named in 22(a). [5 marks]

(c) State five problems of agricultural insurance in West Africa. [5 marks]

(d) Give three disadvantages of television as a method of disseminating agricultural information. [3 marks]

Question 23

(a) Mention three forage legumes that could be found on a rangeland. [3 marks]

(b) List five beekeeping tools. [5 marks]

Question 24

(a) Explain the process of photosynthesis in plants. (6 marks)

(b) List four factors affecting the rate of photosynthesis. (4 marks)

(c) Mention two differences between C3 and C4 plants. (4 marks)

(d) State three ways in which plants utilize the products of photosynthesis. (6 marks)

Question 25

(a) Define soil erosion. (2 marks)

(b) List and explain three types of soil erosion. (6 marks)

(c) State four methods of controlling soil erosion. (4 marks)

(d) Describe how cover cropping helps in controlling soil erosion. (3 marks)

(e) Explain the role of organic matter in soil conservation. (3 marks)

Question 26

(a) Define the term transpiration. (2 marks)

(b) State four factors that affect the rate of transpiration in plants. (4 marks)

(c) Explain three ways in which transpiration is important to plants. (6 marks)

(d) Mention three differences between transpiration and guttation. (3 marks)

(e) Describe the structure and function of stomata. (5 marks)

Question 27

(a) What is a balanced diet? (2 marks)

(b) List and describe five classes of nutrients in a balanced diet. (10 marks)

(c) State four factors that influence the nutritional requirements of farm animals. (4 marks)

(d) Mention four signs of nutritional deficiency in farm animals. (4 marks)

Question 28

(a) Define the term hybridization in crop production. (2 marks)

(b) Explain three types of hybridization. (6 marks)

(c) State four advantages of hybridization in crop improvement. (4 marks)

(d) Mention two challenges associated with hybridization. (2 marks)

(e) Discuss the role of hybridization in plant breeding. (6 marks)

Question 29

(a) What is animal husbandry? (2 marks)

(b) Mention and describe four systems of animal husbandry. (8 marks)

(c) State five advantages of intensive animal farming. (5 marks)

(d) Explain three problems associated with extensive animal farming. (5 marks)

Question 30

- (a) Define the term soil fertility. (2 marks)
- (b) List five factors that affect soil fertility. (5 marks)
- (c) Describe three methods of improving soil fertility. (6 marks)
- (d) Explain the role of legumes in maintaining soil fertility. (4 marks)
- (e) Mention three organic fertilizers and their benefits. (3 marks)

Question 31

- (a) What is selective breeding in animal production? (2 marks)
- (b) Describe three methods of selective breeding. (6 marks)
- (c) State four advantages of selective breeding in livestock production. (4 marks)
- (d) Mention three disadvantages of selective breeding. (3 marks)
- (e) Explain the importance of crossbreeding in livestock improvement. (5 marks)

Question 32

- (a) Define the term irrigation. (2 marks)
- (b) Explain three methods of irrigation used in crop production. (6 marks)
- (c) State four advantages of irrigation in agriculture. (4 marks)
- (d) Mention three challenges faced in the practice of irrigation. (3 marks)
- (e) Discuss the role of irrigation in increasing crop yield. (5 marks)

Question 33

- (a) What is farm mechanization? (2 marks)
- (b) List and describe three types of farm machinery used in agriculture. (6 marks)
- (c) State four benefits of farm mechanization. (4 marks)
- (d) Mention three challenges of farm mechanization in West Africa. (3 marks)
- (e) Explain the impact of farm mechanization on agricultural productivity. (5 marks)

Question 34

- (a) Define the term genetic engineering. (2 marks)
- (b) Describe three techniques used in genetic engineering in agriculture. (6 marks)
- (c) State four benefits of genetic engineering in crop production. (4 marks)
- (d) Mention two ethical concerns associated with genetic engineering. (2 marks)
- (e) Discuss the future prospects of genetic engineering in agriculture. (6 marks)

Question 35

- (a) What is pest control? (2 marks)
- (b) Explain four methods of pest control in agriculture. (8 marks)
- (c) State three advantages and two disadvantages of using chemical pesticides. (5 marks)
- (d) Mention two biological control methods and their benefits. (3 marks)
- (e) Discuss the impact of pests on crop yield. (5 marks)

Question 36

- (a) Define the term crop rotation. (2 marks)
- (b) Describe three benefits of crop rotation in agriculture. (6 marks)

- (c) State four crops suitable for rotation with maize. (4 marks)
- (d) Mention two challenges of practicing crop rotation. (2 marks)
- (e) Explain the role of crop rotation in soil fertility management. (6 marks)

Question 37

- (a) What is agroforestry? (2 marks)
- (b) List and explain four types of agroforestry systems. (8 marks)
- (c) State three advantages of agroforestry in sustainable agriculture. (3 marks)
- (d) Mention two challenges associated with agroforestry. (2 marks)
- (e) Discuss the role of agroforestry in environmental conservation. (5 marks)

Question 38

- (a) Define the term apiculture. (2 marks)
- (b) Describe three types of beehives used in apiculture. (6 marks)
- (c) State four products obtained from beekeeping. (4 marks)
- (d) Mention three benefits of apiculture to the environment. (3 marks)
- (e) Explain the challenges faced in apiculture in tropical regions. (5 marks)

Question 39

- (a) What is sustainable agriculture? (2 marks)
- (b) List and explain four principles of sustainable agriculture. (8 marks)
- (c) State three benefits of practicing sustainable agriculture. (3 marks)
- (d) Mention two challenges faced in implementing sustainable agriculture practices. (2 marks)
- (e) Discuss the role of sustainable agriculture in food security. (5 marks)

Question 40

- (a) Define the term organic farming. (2 marks)
- (b) Explain three principles of organic farming. (6 marks)
- (c) State four advantages of organic farming over conventional farming. (4 marks)
- (d) Mention three challenges of organic farming. (3 marks)
- (e) Discuss the impact of organic farming on soil health. (5 marks)

Question 41

- (a) What is food security? (2 marks)
- (b) List and explain four factors affecting food security in West Africa. (8 marks)
- (c) State three measures to improve food security. (3 marks)
- (d) Mention two challenges of achieving food security in developing countries. (2 marks)
- (e) Discuss the role of government in ensuring food security. (5 marks)

Question 42

- (a) Define the term soil conservation. (2 marks)
- (b) Explain three methods of soil conservation. (6 marks)
- (c) State four benefits of soil conservation practices. (4 marks)
- (d) Mention three challenges of soil conservation in tropical regions. (3 marks)
- (e) Discuss the impact of deforestation on soil conservation. (5 marks)

Question 43

- (a) What is post-harvest technology? (2 marks)
- (b) List and explain three methods of post-harvest handling of crops. (6 marks)
- (c) State four advantages of proper post-harvest handling. (4 marks)
- (d) Mention three challenges faced in post-harvest technology. (3 marks)
- (e) Discuss the role of post-harvest technology in reducing food loss. (5 marks)

Question 44

- (a) Define the term animal nutrition. (2 marks)
- (b) List and describe four essential nutrients required by farm animals. (8 marks)
- (c) State three factors affecting the nutritional requirements of farm animals. (3 marks)
- (d) Mention two challenges of providing adequate nutrition to farm animals. (2 marks)
- (e) Explain the role of nutrition in animal health and productivity. (5 marks)

Question 45

- (a) What is aquaculture? (2 marks)
- (b) List and explain three types of aquaculture systems. (6 marks)
- (c) State four benefits of aquaculture. (4 marks)
- (d) Mention three challenges faced in aquaculture. (3 marks)
- (e) Discuss the role of aquaculture in meeting global fish demand. (5 marks)

Question 46

- (a) Define the term climate change. (2 marks)
- (b) Explain three ways in which climate change affects agriculture. (6 marks)
- (c) State four measures to mitigate the impact of climate change on agriculture. (4 marks)
- (d) Mention three challenges in addressing climate change in agriculture. (3 marks)
- (e) Discuss the role of agricultural practices in contributing to climate change. (5 marks)

Question 47

- (a) What is integrated pest management (IPM)? (2 marks)
- (b) List and explain four components of IPM. (8 marks)
- (c) State three benefits of IPM over conventional pest control methods. (3 marks)
- (d) Mention two challenges of implementing IPM. (2 marks)
- (e) Discuss the role of IPM in sustainable agriculture. (5 marks)

Question 48

- (a) Define the term rural development. (2 marks)
- (b) List and explain three strategies for rural development in agriculture. (6 marks)
- (c) State four benefits of rural development to agriculture. (4 marks)
- (d) Mention three challenges faced in rural development. (3 marks)
- (e) Discuss the impact of rural development on agricultural productivity. (5 marks)

Question 49 (a) State four characteristics of commercial agriculture. (4 marks)

- (b) Mention five governmental programs that assist in agricultural land development in West Africa. (5 marks)
- (c) State three functions of each of the following farm machinery:

- (i) bulldozer;
- (ii) mist blower;
- (iii) tractor. (9 marks)

Question 50 (i) Define the term bush fallowing. (2 marks)

(ii) Enumerate four benefits of bush fallowing in agriculture. (4 marks)

(b) State three effects of each of the following farming practices on soils:

- (i) harrowing;
- (ii) flooding;
- (iii) overgrazing. (9 marks)

(c) List three common pasture grasses. (3 marks)

Question 51 (a) Using the table below, name two agricultural raw materials used by each of the following agro-based industries:

Agro-based Industry	Agricultural Raw Material
Soap industry	(i) _____
	(ii) _____
Textile industry	(i) _____
	(ii) _____
Beverage industry	(i) _____
	(ii) _____
Canning industry	(i) _____
	(ii) _____

(8 marks)

(b) (i) Explain briefly the term nursery as used in crop production. (2 marks)

(ii) List four factors that should be considered in selecting a site for the establishment of a crop nursery. (4 marks)

(c) State four advantages of drip irrigation. (4 marks)

Question 52 (a) Discuss aphids as a crop pest under the following headings:

- (i) three crops attacked;
- (ii) two economic importance;
- (iii) two prevention and control measures. (7 marks)

(b) List five characteristics of farm animals that could be used for selection in animal improvement. (5 marks)

(c) Explain each of the following terms as used in forest management:

- (i) deforestation;
- (ii) taungya farming. (4 marks)
- (d) Define the term horticulture. (2 marks)

Question 53 (a) State three reasons for carrying out each of the following animal husbandry practices:

- (i) culling of poultry;

- (ii) creep feeding of pigs. (6 marks)
- (b) Describe the process of castrating a lamb using the burdizzo. (5 marks)
- (c) State three precautions to be taken when using a herbicide. (3 marks)
- (d) Explain each of the following agricultural extension methods:
 - (i) demonstration;
 - (ii) agricultural shows. (4 marks)

- Question 54 (a) Draw a marketing channel for table eggs. (7 marks)
- (b) Give three ways of stabilizing the process of agricultural produce in West Africa. (3 marks)
 - (c) List two ways each by which land:
 - (i) appreciates;
 - (ii) depreciates. (4 marks)
 - (d) State two deficiency symptoms of each of the following nutrients in crops:
 - (i) potassium;
 - (ii) phosphorus. (4 marks)

- Question 55 (a) Give four reasons why non-governmental organizations are important in agricultural development in West Africa. (4 marks)
- (b) Mention six problems of commercial agriculture in West Africa. (6 marks)
 - (c) (i) Explain the term deforestation. (2 marks)
 - (ii) State six effects of deforestation on the environment. (6 marks)

- Question 56 (a) State three functions each of the following farm implements during tillage operation:
- (i) disc plough;
 - (ii) disc harrow;
 - (iii) ridger. (9 marks)
 - (b) Mention five ways by which soil air is important. (5 marks)
 - (c) Give four examples of the cultural method of controlling insect pests on the farm. (4 marks)

- Question 57 (a) (i) What is organic agriculture? (2 marks)
- (ii) Give five reasons why farmers should practice organic agriculture. (5 marks)
 - (b) State four ways in which cover crops are important in soil management. (4 marks)
 - (c) Using the table below, classify the following crops based on morphology: (7 marks)

- Question 58 (a) Explain each of the following terms as used in crop production:
- (i) staking;
 - (ii) hardening off;
 - (iii) earthing up;
 - (iv) supplying. (8 marks)
 - (b) Calculate the area of the farmland in hectares based on the provided layout. (5 marks)
 - (c) Outline five features of the battery cage system in the management of layers. (5 marks)

- Question 59 (a) Mention three effects of each of the following changes in climatic factors on egg production:
- (i) temperature;
 - (ii) relative humidity;

- (iii) light. (9 marks)
- (b) State five benefits that could be derived by the farmer using a deep litter system for rearing chickens. (5 marks)
- (c) Give four uses of sugarcane. (4 marks)

Question 60

- (a) A grain dryer was purchased at the cost of ₦200,000.00 and sold at ₦50,000.00 after being used for 10 years. Calculate the:
 - (i) total depreciation;
 - (ii) annual depreciation. (5 marks)
- (b) Outline three roles each of the following marketing agents:
 - (i) retailer;
 - (ii) wholesaler;
 - (iii) cooperative society. (9 marks)
- (c) Give four reasons why a farmer should control the weaver birds in a rice field. (4 marks)

Question 61

- (a) Give four reasons why non-governmental organizations are important in agricultural development in West Africa. (4 marks)
- (b) Mention six problems of commercial agriculture in West Africa. (6 marks)
- (c) (i) Explain the term deforestation. (2 marks)
- (ii) State six effects of deforestation on the environment. (6 marks)

Question 62

- (a) State three functions each of the following farm implements during tillage operation:
 - (i) disc plough;
 - (ii) disc harrow;
 - (iii) ridger. (9 marks)
- (b) Mention five ways by which soil air is important. (5 marks)
- (c) Give four examples of the cultural method of controlling insect pests on the farm. (4 marks)

Question 63

- (a) (i) What is organic agriculture? (2 marks)
- (ii) Give five reasons why farmers should practice organic agriculture. (5 marks)
- (b) State four ways in which cover crops are important in soil management. (4 marks)
- (c) Using the table below, classify the following crops based on morphology: (7 marks)

Question 64

(a) Explain each of the following terms as used in crop production:

- (i) staking;
- (ii) hardening off;
- (iii) earthing up;
- (iv) supplying. (8 marks)

(b) Calculate the area of the farmland in hectares based on the provided layout. (5 marks)

(c) Outline five features of the battery cage system in the management of layers. (5 marks)

Question 65

(a) Mention three effects of each of the following changes in climatic factors on egg production:

- (i) temperature;
- (ii) relative humidity;
- (iii) light. (9 marks)

(b) State five benefits that could be derived by the farmer using a deep litter system for rearing chickens. (5 marks)

(c) Give four uses of sugarcane. (4 marks)

Question 66

(a) A grain dryer was purchased at the cost of ₦200,000.00 and sold at ₦50,000.00 after being used for 10 years. Calculate the:

- (i) total depreciation;
- (ii) annual depreciation. (5 marks)

(b) Outline three roles each of the following marketing agents:

- (i) retailer;
- (ii) wholesaler;
- (iii) cooperative society. (9 marks)

(c) Give four reasons why a farmer should control the weaver birds in a rice field. (4 marks)

Question 67

(a) Differentiate between freehold and leasehold land tenure systems. (4 marks)

(b) Highlight three aims of the International Livestock Centre for Africa (ILCA). (3 marks)

(c) Give six reasons why agriculture has not been fully mechanized in West Africa. (6 marks)

(d) Mention five precautions that should be taken when using a knapsack sprayer. (5 marks)

Question 68

(a) Explain four ways in which the following terms are important in crop production:

- (i) soil profile;
- (ii) soil texture. (8 marks)

(b) (i) Differentiate between a forest and a pasture. (2 marks)

(ii) List four pasture management practices in West Africa. (4 marks)

(c) Enumerate four characteristics of sedimentary rocks. (4 marks)

Question 69

- (a) Draw and label the carbon cycle. (7 marks)
- (b) Give three reasons why the carbon cycle is important to crops and animals. (3 marks)
- (c) List four farm structures used in storing harvested crops. (4 marks)
- (d) Mention four effects of lice in poultry production. (4 marks)

Question 70

- (a) State six advantages of organic fertilizers over inorganic fertilizers. (6 marks)
- (b) If the spacing of tomato is 30 cm x 30 cm, determine the population of tomato plants expected on a farmland measuring 75 m x 51 m. (6 marks)
- (c) (i) What are ornamental plants? (2 marks)
- (ii) Mention four maintenance practices that are carried out on ornamental plants. (4 marks)

Question 71

- (a) A farmer observed twisting of neck and head as well as paralysis of the legs among the chickens on a poultry farm.
 - (i) Name the disease that the chickens are most likely infected with. (1 mark)
 - (ii) What is the causative organism of the disease named in 71(a)(i)? (1 mark)
 - (iii) State three factors that could predispose chickens to the disease named in 71(a)(i). (3 marks)
 - (iv) Mention three ways of preventing an outbreak of the disease named in 71(a)(i) on a poultry farm. (3 marks)
- (b) State three functions each of the following organs in farm animals:
 - (i) liver;
 - (ii) kidney. (6 marks)
- (c) Mention four effects of high wind velocity on growing crops. (4 marks)

Question 72

- (a) What is agricultural credit? (2 marks)
- (b) Give six reasons why farmers find it difficult to obtain loans for agricultural production in West Africa. (6 marks)
- (c) Explain each of the following terms as used in agricultural business:
 - (i) farm valuation;
 - (ii) cash record;
 - (iii) balance sheet. (6 marks)
- (d) Mention four uses of poultry eggs to humans. (4 marks)

Question 73

- (a) Explain five reasons why agriculture is regarded as the bedrock of the economy of West Africa. (10 marks)
- (b) Mention five farm operations which require the use of electrical power. (5 marks)
- (c) State three disadvantages of using electrical power for farm operations. (3 marks)

Question 74

- (a) Describe the process of formation of each of the following rock types:
 - (i) igneous rock;
 - (ii) sedimentary rock;
 - (iii) metamorphic rock. (6 marks)
- (b) Outline six factors that should be considered in selecting a crop variety for cultivation. (6 marks)

marks)

(c) Give three reasons why it is best to raise tomato in the nursery before transplanting. (3 marks)

(d) Mention three farm tools used in pruning ornamental plants. (3 marks)

Question 75

(a) Complete the table below on crop diseases:

Diseases	Crop Affected	Causal Organism	Symptom	Two Control Measures
Rosette	(i)	(ii)	(iii)	(iv)
Gummosis	(vi)	(vii)	(viii)	(ix)
	(v)			

(10 marks)

(b) Give two examples each of the following crops:

(i) grain legumes;

(ii) spices. (4 marks)

(c) State four problems associated with irrigation in West Africa. (4 marks)

Question 76

1. (a) State five reasons why government should enact laws to regulate the use of land for agriculture. [5 marks]
2. (b) A cocoa farmer has just acquired 10 hectares of farmland and is considering the type of farm power to use for tillage operations. (i) Advise the farmer on the most appropriate type of farm power to use. [1 mark] (ii) State four reasons for the answer in 1(b)(i). [4 marks]
3. (c) Mention one function each of the following parts of a seed planter: (i) seed tube; (ii) furrow opener; (iii) hopper; (iv) seed metering device; (v) furrow wheel. [5 marks]
4. (d) List three machines used in processing crops after harvesting. [3 marks]

Question 77

1. (a) Explain each of the following cropping systems:
 1. *monocropping*;
 2. *mixed cropping*;
 3. *mixed farming*;
 4. *crop rotation*. [8 marks]
2. (b) A crop farmer observed that the leaves of the orange plants on the farm turned purple with most of the immature fruits dropping.
 1. Identify the nutrient deficient in the orange plants. [1 mark]
 2. List four inorganic fertilizers that could correct the nutrient deficiency observed in the orange plants. [4 marks]
3. (c) Name five materials that could be used to attract bees to a new beehive. [5 marks]

Question 78

1. (a) Explain four causes of damage to maize during storage. [8 marks]
2. (b) Two heterozygous tall plants, Tt were crossed in an experiment. Using a genetic diagram,
 1. show all the possible offspring that could result from this cross. [2 marks]
 2. determine the:
 1. genotypic ratio of the offspring;
 2. phenotypic ratio of the offspring. [2 marks]
3. (c) State Mendel's first and second laws of inheritance. [4 marks]
4. (d) Mention two forage grasses. [2 marks]

Question 79

- (a) Complete the table below. [9 marks]

<i>Crop</i>	<i>Botanical name</i>	<i>Classification based on use</i>	<i>Method of propagation</i>
Onion	<i>Allium cepa</i>	- Vegetable - Spice	- Seed - Sett - Bulb
Ginger	<i>Zingiber officinale</i>	- Spice	- Rhizome
Jute	<i>Corchorus olitorius</i>	- Fibre - Vegetable	- Seed

- (b) A poultry farmer plans to rear broilers in a deep litter house measuring 36 m long and 10 m wide. If a floor space of 0.36 m² per adult chicken is recommended for broiler production, calculate the number of broilers that the farmer can rear. [4 marks]

- (c) State five advantages of introducing legumes in a pasture. [5 marks]

Question 80

(a) Explain each of the following terms in animal nutrition:

1. *balanced diet*;
2. *maintenance ration*;
3. *production ration*;
4. *malnutrition*. [8 marks]

(b) State six effects of malnutrition on farm animals. [6 marks]

(c) Mention four nursery practices in the production of tomato. [4 marks]

Question 81

1. (a) State four differences between *freehold* and *leasehold* land tenure systems. (4 marks)
2. (b) Highlight three aims of the International Livestock Centre for Africa (*ILCA*). (3 marks)
3. (c) Give six reasons why agriculture has not been fully mechanized in West Africa. (6 marks)
4. (d) Mention five precautions that should be taken when using a knapsack sprayer. (5 marks)

Question 82

1. (a) Explain four ways each in which the following terms are important in crop production:
 - (i) *Soil profile* (8 marks)
 - (ii) *Soil texture* (8 marks)

2. (b)
 - (i) Differentiate between a *forest* and a *pasture*. (2 marks)
 - (ii) List four pasture management practices in West Africa. (4 marks)
3. (c) Enumerate four characteristics of sedimentary rocks. (4 marks)

Question 83

1. (a) Draw and label the carbon cycle. (7 marks)
2. (b) Give three reasons why the carbon cycle is important to crops and animals. (3 marks)
3. (c) List four farm structures used in storing harvested crops. (4 marks)
4. (d) Mention four effects of lice in poultry production. (4 marks)

Question 84

1. (a) State six advantages of organic fertilizers over inorganic fertilizers. (6 marks)
2. (b) If the spacing of tomato is 30 cm x 30 cm, determine the population of tomato plants expected on a farmland measuring 75 m x 51 m. (6 marks)
3. (c) (i) What are *ornamental plants*? (2 marks) (ii) Mention four maintenance practices that are carried out on ornamental plants. (4 marks)

Question 85

(a) A farmer observed twisting of neck and head as well as paralysis of the legs among the chickens on a poultry farm.

- (i) Name the disease that the chickens are most likely infected with. (1 mark)
- (ii) What is the causative organism of the disease named in 5(a)(i)? (1 mark)
- (iii) State three factors that could predispose chickens to the disease named in 5(a)(i). (3 marks)
- (iv) Mention three ways of preventing an outbreak of the disease named in 5(a)(i) on a poultry farm. (3 marks)

(b) State three functions each of the following organs in farm animals:

- (i) liver; (3 marks)
- (ii) kidney. (3 marks)

(c) Mention four effects of high wind velocity on growing crops. (4 marks)

Question 86

1. (a) What is *agricultural credit*? (2 marks)

2. (b) Give six reasons why farmers find it difficult to obtain loans for agricultural production in West Africa. (6 marks)
3. (c) Explain each of the following terms as used in agricultural business:
 - (i) *farm valuation* (6 marks)
 - (ii) *cash record*
 - (iii) *balance sheet*
4. (d) Mention four uses of poultry eggs to humans. (4 marks)

Question 87

- (i) Define agriculture. [2 marks]
- (ii) List six branches of agriculture. [3 marks]
- (a) State five differences between commercial agriculture and subsistence agriculture. [5 marks]
- (b) State three merits and three demerits of using animal power on the farm. [3 marks]

Question 88

- (a) (i) Explain the term farm surveying. [2 marks]
- (ii) State four reasons why farm surveying is important. [4 marks]
- (b) Enumerate six problems facing agricultural development in West Africa. [6 marks]
- (c) List four factors that may affect the efficiency of draught animals. [4 marks]

Question 89

- (a) Define soil erosion. [2 marks]
- (b) List four factors that influence nutrient availability in the soil. [4 marks]
- (c) Explain two roles of each of the following factors in soil formation: [6 marks]
 - (i) Climate;
 - (ii) Living organisms;
 - (iii) Topography.
- (d) State four problems associated with overhead irrigation. [4 marks]

Question 90

- (a) (i) Define crop rotation. [2 marks]
- (ii) Design a four-year crop rotation system using yam, cassava, maize, and groundnut. [8 marks]
- (b) State four reasons why weeds are difficult to control. [4 marks]
- (c) Mention two effects of diseases on crop production. [2 marks]

Question 91

- (a) List four methods of identification in cattle production. [4 marks]

(b) State three reasons for carrying out each of the following animal husbandry practices:

- (i) Culling of poultry;
- (ii) Dehorning of cattle.

(c) Mention the botanical names of four pasture legumes grown in West Africa. [4 marks]

(d) State four ways of preserving harvested fish. [4 marks]

Question 92

(a) Define:

- (i) Demand;
- (ii) Price elasticity of demand. [4 marks]

(b) Explain why the demand curve slopes downwards. [2 marks]

(c) List four agents of agricultural marketing. [4 marks]

(d) State two advantages and one disadvantage of each of the following agricultural extension teaching methods:

- (i) Newspapers;
- (ii) Television. [6 marks]

Question 93

(a) List four factors that cause a change in the supply of agricultural produce. [4 marks]

(b) Distinguish between fixed cost and variable cost, giving two examples in each case. [6 marks]

(c) Explain the following agricultural extension teaching methods:

- (i) Posters;
- (ii) Field trips. [6 marks]

Question 94

(a) Explain the term "sustainable agriculture." [2 marks]

(b) List four benefits of sustainable agriculture. [4 marks]

(c) Describe the concept of integrated pest management (IPM). [4 marks]

Question 95

(a) State three advantages of using organic pesticides. [3 marks]

(b) List three challenges associated with organic farming. [3 marks]

(c) Explain the importance of crop rotation in soil fertility management. [4 marks]

Question 96

(a) Define "precision agriculture." [2 marks]

(b) List four technologies used in precision agriculture. [4 marks]

(c) Describe how precision agriculture can improve crop yields. [4 marks]

Question 97

(a) Differentiate between land ownership by leasehold and land ownership by freehold. [2 marks]

(b) Mention five ways in which non-governmental organizations contribute to agricultural development in West Africa. [5 marks]

(c) State four ways in which rural-urban migration has hindered agricultural development in West Africa. [4 marks]

(d) Mention three socio-economic uses of land. [3 marks]

(e) Explain two ways by which each of the following environmental factors affect agricultural production:

- Soil organisms
- Soil texture [4 marks]

Question 98

(a) State two advantages and two disadvantages of using biogas as a source of farm power. [4 marks]

(b) Mention eight simple farm tools. [4 marks]

(c) List two common storage structures and two processing structures that could be found on a farmstead in West Africa. [4 marks]

(d) Explain the term denitrification. [2 marks]

(e) Describe the following components of the soil:

- Mineral matter
- Organic matter [4 marks]

Question 99

(a) Describe the cultivation of Cowpea under the following headings:

- Soil requirement
- Method of propagation
- Spacing [3 marks]

(b) Name one disease and two field pests of Cowpea. [3 marks]

(c) State four cultural methods of controlling crop pests. [4 marks]

(d) State four human activities that promote deforestation in West Africa. [4 marks]

(e) Describe each of the following agro-forestry practices:

- Alley cropping
- Ley farming [4 marks]

Question 100

(a) In an incubator, 1,500 eggs were arranged on the trays for incubation. During candling, 25 eggs were discovered to be unfertilized. At hatching, 475 chicks were male, and the remaining were female. Calculate the percentage of:

- Unfertilized eggs
- Fertilized eggs
- Male chicks
- Female chicks [6 marks]

(b) State four signs that indicate the approach of parturition in a cow. [4 marks]

(c) Mention four vegetative materials that could be used in propagating ornamental plants. [4 marks]

(d) State four advantages of grafting. [4 marks]

Question 101

(a) State four reasons why the prices of most agricultural produce are unstable in West Africa. [4 marks]

(b) Explain the following agricultural extension methods:

- Method demonstration
- Agricultural shows [4 marks]

(c) Discuss the Cercospora leaf spot disease under the following headings:

- Crop affected
- Causative agent
- Mode of transmission
- Two symptoms
- One preventive measure [6 marks]

(d) State four advantages of selection as a method of crop improvement. [4 marks]

Question 102

(a) State five roles of agricultural research institutes in agricultural development. [5 marks]

(b) Outline two advantages and two disadvantages of land ownership by inheritance. [4 marks]

(c) Mention four principles of farmstead layout. [4 marks]

(d) What is soil profile? [2 marks]

(e) (i) State two uses of the knapsack sprayer. (ii) List three secondary tillage implements. [5 marks]

Question 103

(a) Define the term soil pH. [2 marks]

(b) List three materials used in correcting soil acidity. [3 marks]

(c) Using a table, classify the following crops based on their uses:

- Cowpea, Amaranthus, Oil palm, Coconut, Cassava, Maize, Soya bean, Cocoa, Pepper, and Plantain [5 marks]

(d) Give the botanical name of each of the following crops:

- Cowpea
- Rubber
- Oil palm
- Yam
- Cocoa [5 marks]

Question 104

(a) Discuss the Onion Twister disease under the following headings:

- Causative organism
- Mode of transmission
- One symptom
- Two control measures [5 marks]

(b) State five adverse effects of insect pests on crop production. [5 marks]

(c) Outline four desirable qualities to be considered in selecting a breeding stock in pig production. [4 marks]

(d) List four characteristics of beef cattle. [4 marks]

Question 105

(a) State three reasons why livestock distribution is concentrated in the savannah region of West Africa. [3 marks]

(b) If the recommended floor space for White Leghorn fowl in a deep litter house is three fowls per square meter, calculate the number of White Leghorn fowls that can be kept in a deep litter house measuring 20m x 100m. [3 marks]

(c) What is artificial insemination? [2 marks]

(d) State four advantages of artificial insemination. [4 marks]

(e) What is floriculture? [2 marks]

(f) List four ornamental plants used as hedges in West Africa. [4 marks]

Question 106

(a) State three advantages and three disadvantages of sprinkler irrigation. [6 marks]

(b) Outline four ways in which irrigation is important in crop production. [4 marks]

(c) What is a farm budget? [2 marks]

(d) A tractor was bought for D200,000.00 in 1996 and sold for D30,000.00 in 2006 when it was no longer serviceable. Determine the:

- Salvage value of the tractor
- Useful life of the tractor
- Total depreciation of the tractor
- Annual depreciation of the tractor [4 marks]